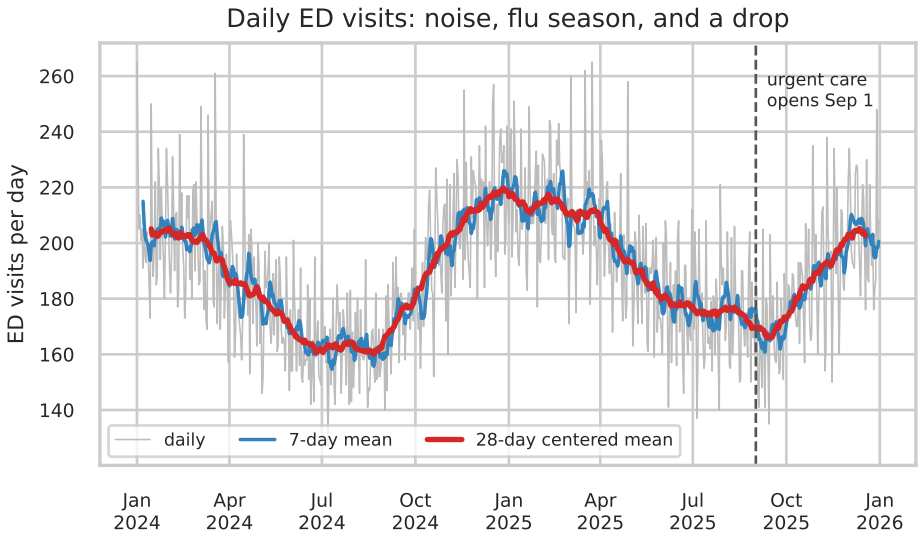
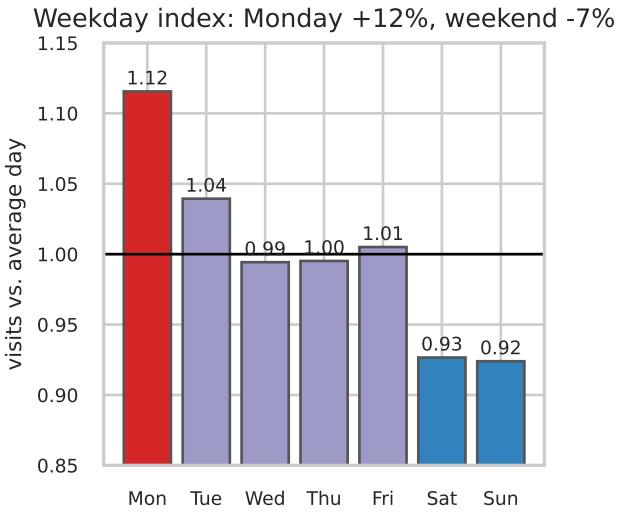
Day 10 Cheat Sheet: Time Series EDA

Daily ED visits: rolling means, seasonality, and real shifts vs. bumpy weeks

TERM	WHAT IT MEANS	CODE
DatetimeIndex	dates as the index: slicing, resampling, lags	`read_csv(parse_dates=[..]).set_index(..)`
asfreq("D")	re-insert missing days as NaN, then fill	`s.asfreq("D").interpolate()`
Rolling mean	average of the last k days (7 cancels weekdays)	`s.rolling(7).mean()`
Resample	change grain: daily to weekly / monthly	`s.resample("W-SUN").sum()`
Seasonality	a repeating pattern (weekday, flu season)	`s.groupby(s.index.dayofweek).mean()`
Autocorrelation	how much today looks like k days ago	`s.autocorr(7)`
STL decomposition	trend + seasonal + remainder	`STL(s, period=7, robust=True).fit()`
Year over year	same period last year: cancels season	`s.pct_change(12)` on monthly data
Control chart	center +/- 3 SD from a stable baseline	flag points outside, and long runs
Lag feature	yesterday / last week as a model input	`s.shift(7)`; never shuffle time





THIS DATA (simulated): 731 days (Jan 2024 to Dec 2025), 9 days missing in the extract, filled by interpolation
Mean 189.2 visits/day (SD 25.0); winter months ~217/day vs summer ~161/day
Weekly pattern: Monday index 1.12, Sat/Sun 0.93; lag-7 autocorrelation 0.70 vs lag-1 0.51
Jan-Aug 2025 ran about +7% vs 2024; Sep-Dec 2025 ran -5.1% (Welch p = 0.001)
YoY control chart: 9 of 17 weeks after Sep 1 below the LCL, all 17 below center (run rule fires)

HABITS THAT KEEP YOU HONEST
- Check for missing dates first: a gap looks like a drop on a chart
- Compare like with like: same weekday, same season, same week last year
- One odd week is noise; a run of 8+ on one side of the center is a shift
- Split by time (train on the past, test on the future), never at random
- Build lag and rolling features with shift() first, so today never leaks in

```
ed = pd.read_csv("ed_visits.csv", parse_dates=["visit_date"]).set_index("visit_date")
full = ed.asfreq("D"); full["visits"] = full["visits"].interpolate()
full["ma7"] = full["visits"].rolling(7).mean();  weekly = full["visits"].resample("W-SUN").sum()
ratio = weekly / weekly.shift(52)        # this week vs. the same week last year
```

Infographic: Is That Drop Real?

Simulated: 731 days of emergency department visits at one hospital

Load	Gaps	Smooth	Season	Compare	Decide
parse dates, set index	asfreq('D'), interpolate	7 / 28-day rolling mean	weekday + month profile	same week last year	limits + run rule

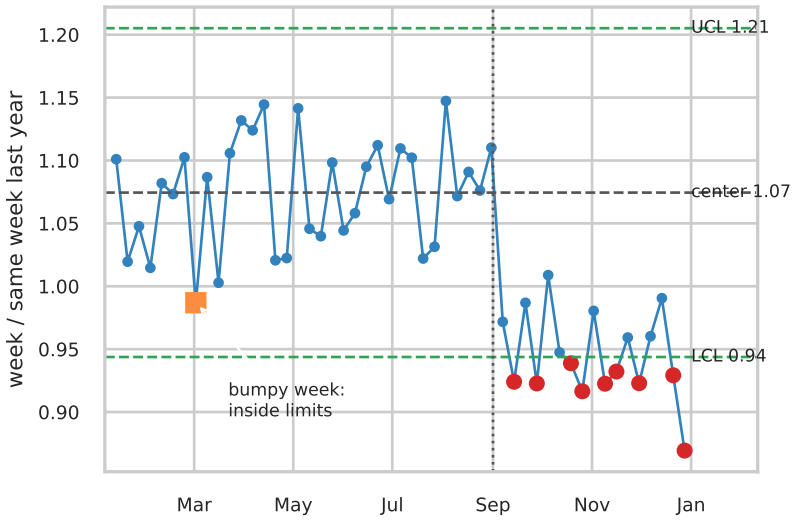
MONDAY PEAK
+12%
vs average day

LAG-7 ACF
0.70
weekly rhythm

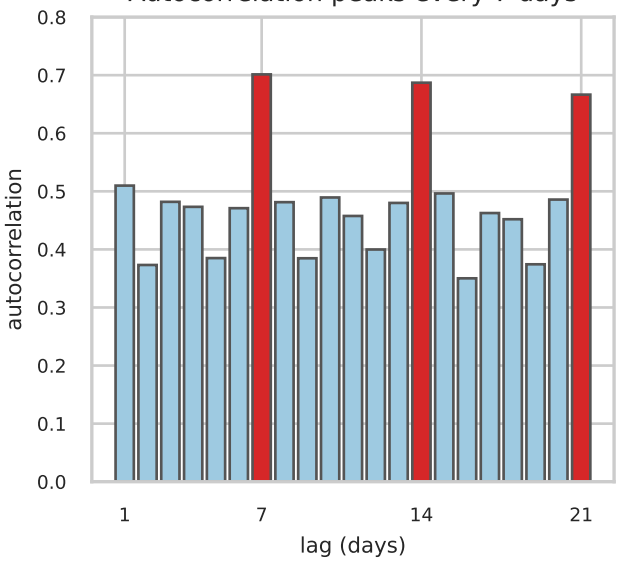
SEP-DEC YoY
-5.1%
was +7%

WEEKS < LCL
9 of 17
after Sep 1

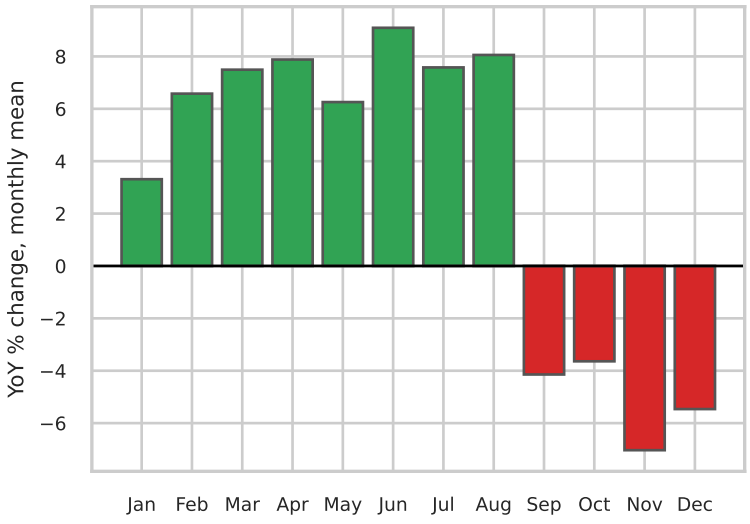
YoY control chart: 9 weeks below LCL, 17-week run



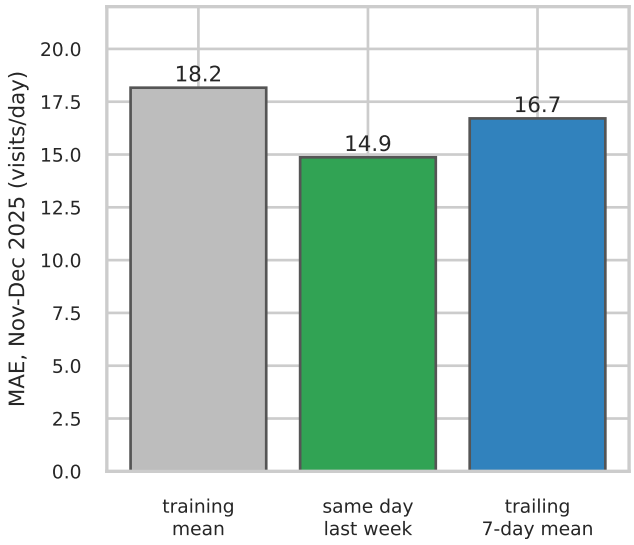
Autocorrelation peaks every 7 days



2025 vs 2024 by month: +7% flips to -5%



Baselines to beat (lower is better)



Read it like a manager: ED volume fell about 12 points against last year starting in September, right when the new urgent care opened. That is a sustained shift, not a bumpy week. The timing fits, but this chart alone does not prove the cause. Simulated data.

Hands-On Worksheet: ds-genai-90-days / day10

60-90 minutes | notebook: day10/ed_time_series.ipynb

[ ] mkdir day10, conda activate ds90, open ed_time_series.ipynb

[ ] Simulate ed_visits.csv with seed 10 (snippet 1); 722 rows, 9 gaps

[ ] parse_dates + set_index; asfreq('D') and interpolate the 9 gaps

[ ] Plot daily visits with 7-day and 28-day rolling means

[ ] Resample weekly (W-SUN) and monthly; which months are busiest?

[ ] Weekday index: Monday vs. Sunday. autocorr at lags 1, 7, 14

[ ] STL(period=7): plot trend, seasonal, remainder

[ ] Monthly YoY % change: when does it flip negative?

[ ] Weekly YoY ratio + control limits from Jan-Aug 2025 (snippet 4)

[ ] Count weeks below LCL; find the run of 8+ below center

[ ] Lag features with shift(); time split at Nov 1, 2025

[ ] MAE of 3 baselines: mean, same day last week, 7-day mean

[ ] Findings cell; git add day10 && git commit && git push

[ ] Stretch: Snowflake 7-day moving average with a window function

```
import pandas as pd
from statsmodels.tsa.seasonal import STL
ed = pd.read_csv("ed_visits.csv", parse_dates=["visit_date"]).set_index("visit_date")
full = ed.asfreq("D"); full["visits"] = full["visits"].interpolate()
print(full["visits"].autocorr(7))                          # expect ~0.70
weekly = full["visits"].resample("W-SUN").sum().iloc[1:-1]
ratio = (weekly / weekly.shift(52)).dropna()               # YoY, same week
stl = STL(full["visits"], period=7, robust=True).fit(); stl.plot()
```

Missing days: ____ Mean/day: _____ Monday index: _____ lag-7 ACF: _____ Busiest month: _____

Baseline ratio: _____ LCL: _____ Weeks < LCL: _____ MAE (last week): _____ MAE (mean): _____

Day 10 Self-Check Quiz

1. Why call asfreq('D') before plotting or computing rolling means?

   _____

2. Why does a 7-day rolling mean look so much smoother than the daily series?

   _____

3. Lag-7 autocorrelation is 0.70 but lag-1 is 0.51. What does that tell you?

   _____

4. Why compare Sep-Dec 2025 with Sep-Dec 2024 instead of with Jan-Aug 2025?

   _____

5. One baseline week dips well below center but stays inside the limits. Act?

   _____

6. All 17 weeks after Sep 1 sit below the center line. Why is that a signal?

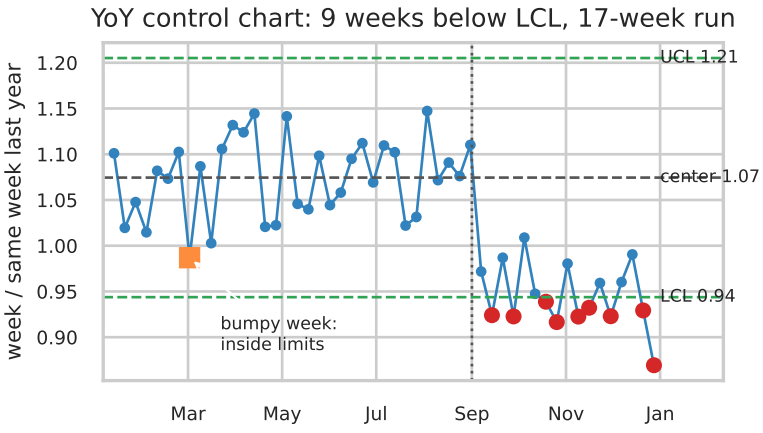
   _____

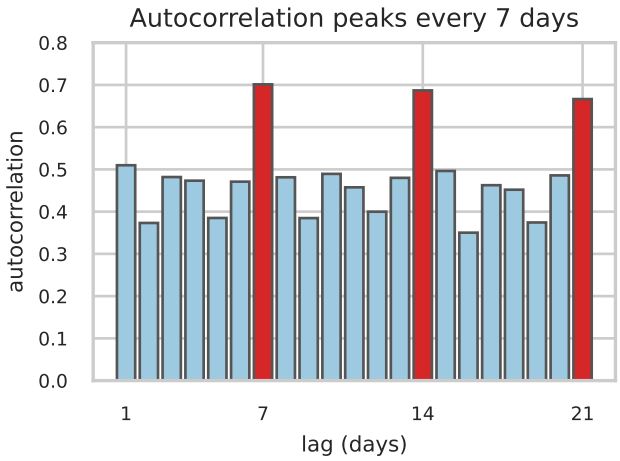
7. What goes wrong if you shuffle rows before a train/test split here?

   _____

8. The urgent care opened Sep 1 and volume fell. Can you claim it caused the drop?

   _____





Answers:
1 missing days silently vanish from a daily series; asfreq exposes them as NaN so a gap is not mistaken for a drop or a shorter window
2 every 7-day window holds exactly one of each weekday, so the weekday cycle cancels and only trend, season and real changes remain
3 the strongest memory is weekly: a Monday looks most like last Monday, so weekday matters more than yesterday's count
4 Jan-Aug includes flu season and the weekday mix differs; same months last year cancel season, so the change is easier to trust
5 no: a single point inside the 3-SD limits is ordinary variation (a bumpy week). Note it and keep watching
6 by chance each week is ~50/50 above or below; 17 in a row is extremely unlikely (8 in a row is the usual run rule), so the level shifted
7 future days leak into training (lags and rolling means overlap the test period), so the error looks better than it will be in real use
8 not on its own: the timing fits, but this is observational. Check other causes (coding change, closures, a mild flu season) first